Lagrange Interpolation Problem & Numerical Integration

Numerical Integration

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#Empowering Minds.

Plan

Numerical Integration

References

- **Main reference:**
  - Richard L. Burden & J. Douglas Faires, *Numerical Analysis*, 9^{th} Edition, Brooks/Cole, Cengage Learning, 2011.
- **Advanced reference (theoretical aspects):**
  - J. Stoer & R. Bulirsch, *Introduction to Numerical Analysis*, Texts in Applied Mathematics 12, 3^{rd} Edition, Springer, 2002.

Introduction

In many applications, we need to ***compute integrals*** of the form

$$\int_a^b f(x)\,dx.$$

- Sometimes, the function f is ***not known explicitly***
- Or the integral ***cannot*** be computed ***exactly***; like the case $\int_0^1 e^{-x^2}\,dx$.
- Only ***discrete values*** of f are available

Objective

Approximate the ***integral*** using a ***finite number of function evaluations***

- ***Numerical integration*** allows us to handle ***complex or unknown functions***

SIMPLE RULES

Numerical Integration

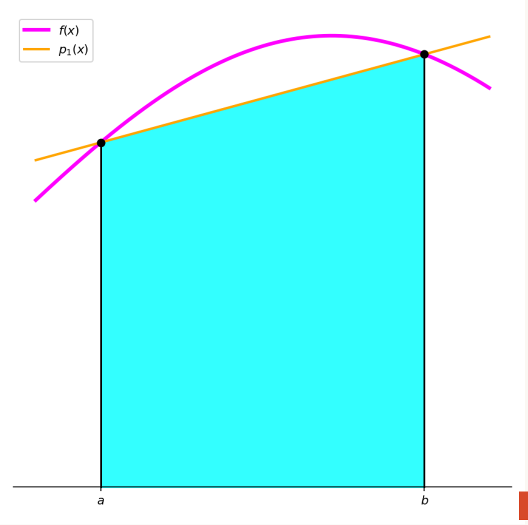
The Trapezoidal Rule: Idea

Approximate $f : [a, b] \to \mathbb{R}$ by a ***linear interpolation*** on the points $(a, f(a))$ and $(b, f(b))$. The ***interpolating polynomial*** is given by

$$P_1(x) = f(a)\frac{x-b}{a-b} + f(b)\frac{x-a}{b-a}.$$

We replace f by P_1 in the ***integral***

$$I(f) \approx I(P_1) = \int_a^b P_1(x)\,dx.$$



Numerical Integration

The Trapezoidal Rule

Let us compute the ***integral of the interpolating polynomial*** $I(P_1)$. Using the expression of $P_1(x)$, we obtain

$$I(P_1) = f(a)\int_a^b \frac{x-b}{a-b}\,dx + f(b)\int_a^b \frac{x-a}{b-a}\,dx.$$

A direct computation gives

$$I(P_1) = \frac{b-a}{2}\bigl(f(a)+f(b)\bigr).$$

Trapezoidal Rule

$$\int_a^b f(x)\,dx \approx \frac{b-a}{2}\bigl(f(a)+f(b)\bigr).$$

Numerical Integration

The Trapezoidal Rule: Example 1

Consider the integral

$$I = \int_0^{\frac{\pi}{2}} \sin(x)\, dx.$$

We apply the ***Trapezoidal Rule***: Here, $a = 0$, $b = \frac{\pi}{2}$:

$$I \approx \frac{\pi/2}{2}\left(\sin(0) + \sin(\tfrac{\pi}{2})\right) = \frac{\pi}{4}.$$

Exact value: 1 Approximation: $\frac{\pi}{4} \approx 0.78539816$ Error: $\left|1 - \frac{\pi}{4}\right| \approx 0.21460184$

Consider the integral

$$I = \int_0^{\frac{1}{2}} e^x \sin x \, dx.$$

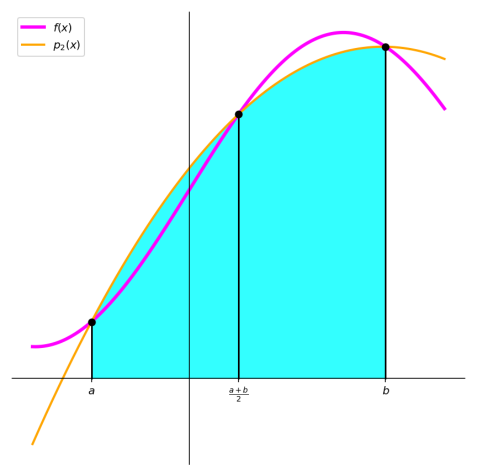
We apply the ***Trapezoidal Rule***:

$$I \approx \frac{1/2}{2}\left(f(0) + f(1/2)\right) \approx 0.1972169.$$

Exact : 0.1716406 *Approx* : 0.1972169 *Error* : 0.0255763

Numerical Integration

Simpson's Rule: Idea



Approximate $f : [a, b] \to \mathbb{R}$ by a ***quadratic interpolation*** on the points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$ with

$$x_0 = a, \quad x_2 = b, \quad x_1 = \frac{a+b}{2}.$$

The ***interpolating polynomial*** is given by the ***Lagrange form***

$$P_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x).$$

We replace f by P_2 in the ***integral***

$$I(f) \approx I(P_2) = \int_a^b P_2(x)\,dx.$$

Numerical Integration

Simpson's Rule: Computation of $I(P_2)$

Let $x_0 = a$, $x_2 = b$, and $x_1 = \dfrac{a+b}{2}$.

We write the ***quadratic interpolant*** in Lagrange form:

$$P_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x),$$

where

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}, \quad L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}, \quad L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}.$$

We compute

$$I(P_2) = \int_a^b P_2(x)\,dx = f(x_0)\int_a^b L_0(x)\,dx + f(x_1)\int_a^b L_1(x)\,dx + f(x_2)\int_a^b L_2(x)\,dx.$$

Let us compute the ***integral of the interpolating polynomial*** $I(P_2)$. We recall

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}, \quad L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}, \quad L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}.$$

We compute

$$\int_a^b L_0(x)\,dx = \frac{b-a}{6}, \quad \int_a^b L_1(x)\,dx = \frac{2(b-a)}{3}, \quad \int_a^b L_2(x)\,dx = \frac{b-a}{6}.$$

Thus

$$\begin{aligned} I(P_2) = \int_a^b P_2(x)\,dx &= f(x_0)\int_a^b L_0(x)\,dx + f(x_1)\int_a^b L_1(x)\,dx + f(x_2)\int_a^b L_2(x)\,dx \\ &= f(x_0)\frac{b-a}{6} + f(x_1)\frac{2(b-a)}{3} + f(x_2)\frac{b-a}{6} \end{aligned}$$

This leads to the following ***formula***

$$I(P_2) = \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right).$$

Simpson's Rule

$$\int_a^b f(x)\,dx \approx \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right).$$

Numerical Integration

Simpson's Rule: Example 1

Consider the integral

$$I = \int_0^{\frac{\pi}{2}} \sin(x)\, dx.$$

We apply ***Simpson's Rule***: Here, $a = 0$, $b = \frac{\pi}{2}$, $\frac{a+b}{2} = \frac{\pi}{4}$:

$$I \approx \frac{\pi/2}{6}\left(\sin(0) + 4\sin(\tfrac{\pi}{4}) + \sin(\tfrac{\pi}{2})\right) = \frac{\pi}{12}\left(0 + 4\cdot\frac{\sqrt{2}}{2} + 1\right) = \frac{\pi}{12}(2\sqrt{2}+1).$$

Comparison

$$\text{Exact value: } 1 \qquad \text{Approximation: } \frac{\pi}{12}(2\sqrt{2}+1) \approx 1.00228$$

$$\text{Error: } \left|1 - \frac{\pi}{12}(2\sqrt{2}+1)\right| \approx 0.00228$$

Consider the integral

$$I = \int_0^{\frac{1}{2}} e^x \sin x \, dx.$$

We apply ***Simpson's Rule***:

$$I \approx \frac{1/2}{6}\Big(f(0) + 4f(1/4) + f(1/2)\Big) \approx 0.1719657.$$

$Exact : 0.1716406 \qquad Approx : 0.1719657 \qquad Error : 0.0003251$

Now we describe the ***general case***. We consider $n+1$ ***equidistant points*** in $[a, b]$:

$$a = x_0 < x_1 < \cdots < x_n = b, \quad x_i = a + ih, \quad h = \frac{b-a}{n}.$$

We approximate $f : [a, b] \to \mathbb{R}$ by a ***polynomial interpolation*** of degree n on these points. The ***interpolating polynomial*** is given by

$$P_n(x) = \sum_{j=0}^{n} f(x_j) L_j(x).$$

We replace f by P_n in the ***integral***:

$$\int_a^b f(x)\,dx \approx \int_a^b P_n(x)\,dx.$$

Using the ***Lagrange representation***, we obtain

$$\int_a^b f(x)\,dx \approx \sum_{j=0}^{n} f(x_j) \int_a^b L_j(x)\,dx.$$

We define the ***weights***

$$w_j = \int_a^b L_j(x)\,dx, \quad j = 0, \ldots, n.$$

Thus, the ***approximation*** reads

Newton–Cotes Formula

$$\int_a^b f(x)\,dx \approx \sum_{j=0}^{n} w_j\, f(x_j).$$

The weights w_j depend only on n and can be computed explicitly.

- $n = 1$ Trapezoidal rule,
- $n = 2$ Simpson rule.

These formulas are obtained by integrating ***polynomial interpolants***

They provide ***explicit approximations*** of the integral

We consider $n = 3$ with ***equidistant nodes***

$$x_0 = a, \quad x_1 = a + h, \quad x_2 = a + 2h, \quad x_3 = a + 3h = b, \quad h = \frac{b-a}{3}.$$

We approximate f by a ***cubic interpolation***

$$P_3(x) = \sum_{j=0}^{3} f(x_j) L_j(x).$$

We replace f by P_3 in the ***integral***

$$\int_a^b f(x)\,dx \approx \int_a^b P_3(x)\,dx = \sum_{j=0}^{3} f(x_j) \int_a^b L_j(x)\,dx.$$

Using the ***change of variable***

$$x = a + ht, \quad t \in [0, 3],$$

we obtain

$$\int_a^b L_j(x)\,dx = h \int_0^3 \ell_j(t)\,dt,$$

where ℓ_j are the ***Lagrange polynomials*** associated with $t = 0, 1, 2, 3$. Hence,

$$\ell_0(t) = \frac{(t-1)(t-2)(t-3)}{(0-1)(0-2)(0-3)} = -\frac{(t-1)(t-2)(t-3)}{6}.$$

A direct computation gives

$$\int_0^3 \ell_0(t)\,dt = \frac{3}{8}.$$

Similarly, one obtains

$$\int_0^3 \ell_1(t)\,dt = \int_0^3 \ell_2(t)\,dt = \frac{9}{8}, \quad \int_0^3 \ell_3(t)\,dt = \frac{3}{8}.$$

Numerical Integration

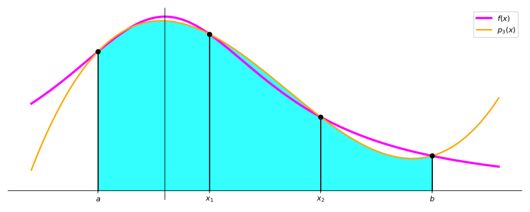
Newton–Cotes Formulas: Simpson 3/8 Rule

Thus, the ***weights*** are

$$w_0 = w_3 = \frac{3h}{8}, \quad w_1 = w_2 = \frac{9h}{8}.$$

Simpson 3/8 **Rule**

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\Big(f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\Big).$$



The ***weights*** of Newton–Cotes formulas are ***precomputed and tabulated***. To do so, let $n \in \mathbb{N}$ and consider the ***Newton–Cotes formula***

$$\int_a^b f(x)\,dx \approx \int_a^b P_n(x)\,dx = h\sum_{i=0}^{n} \alpha_i f(x_i), \quad f(x_i) = f(a+ih).$$

The coefficients α_i are ***rational numbers***. If s is a common denominator of the weights α_i, we define

$$\sigma_i = s\,\alpha_i, \quad i = 0,\dots,n,$$

so that the σ_i are integers. Thus, the formula becomes

$$\int_a^b f(x)\,dx \approx \frac{h}{s}\sum_{i=0}^{n} \sigma_i f(x_i).$$

👉 The factor s and the coefficients σ_i are ***tabulated*** and ***characterize the metho***

Numerical Integration

Simpson 3/8 Rule: Example 1

Consider the integral

$$I = \int_0^{\frac{\pi}{2}} \sin(x)\,dx.$$

We apply ***Simpson* 3/8 *Rule***:

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\Big(f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\Big), \quad h = \frac{b-a}{3}.$$

Here, $x_0 = 0, \quad x_1 = \frac{\pi}{6}, \quad x_2 = \frac{\pi}{3}, \quad x_3 = \frac{\pi}{2}.$

$$I \approx \frac{\pi}{16}\Big(\sin(0) + 3\sin(\tfrac{\pi}{6}) + 3\sin(\tfrac{\pi}{3}) + \sin(\tfrac{\pi}{2})\Big) \approx 1.0016.$$

Exact value: 1 Approximation: ≈ 1.001 Error: ≈ 0.0016

Numerical Integration

Simpson 3/8 Rule: Example 2

Consider the integral

$$I = \int_0^{\frac{1}{2}} e^x \sin x \, dx.$$

We apply ***Simpson*** 3/8 ***Rule*** with $h = \frac{b-a}{3} = \frac{1}{6}$, $x_0 = 0$, $x_1 = \frac{1}{6}$, $x_2 = \frac{1}{3}$, $x_3 = \frac{1}{2}$. Thus

$$\begin{aligned} I &\approx \frac{3h}{8}\Big(f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\Big) \\ &= \frac{1}{16}\Big(0 + 3e^{1/6}\sin(\tfrac{1}{6}) + 3e^{1/3}\sin(\tfrac{1}{3}) + e^{1/2}\sin(\tfrac{1}{2})\Big) \\ &\approx 0.1717693. \end{aligned}$$

Exact : 0.1716406 *Approx* : 0.1717693 *Error* : 0.0001287

Numerical Integration

Newton–Cotes Formulas

n	σ_i	ns	Name
1	1 1	2	Trapezoidal rule
2	1 4 1	6	Simpson's rule
3	1 3 3 1	8	Simpson 3/8 rule
4	7 32 12 32 7	90	Milne's rule
5	19 75 50 50 75 19	288	—
6	41 216 27 272 27 216 41	840	Weddle's rule

Numerical Integration

Newton–Cotes Formulas

- Newton–Cotes formulas provide **simple** and **explicit formulas** to approximate integrals on $[a, b]$
- They are based on **polynomial interpolation**:
  - degree 1 $\rightarrow$ **Trapezoidal rule**
  - degree 2 $\rightarrow$ **Simpson's rule**
  - higher degrees $\rightarrow$ **higher-order formulas**

☞ These approximations are accurate if the interval $[a, b]$ is **small**.

☞ For large intervals, the approximation may be **poor**.

☞ Idea: split $[a, b]$ into **smaller subintervals** and apply the formulas locally.

☞ This leads to the **composite Newton–Cotes formulas**. We describe **Composite Trapezoidal** and **Composite Simpson Rules**.

Composite Rules

We subdivide the interval $[a, b]$ into n ***subintervals***:

$$a = x_0 < x_1 < \cdots < x_n = b, \quad x_i = a + ih, \quad h = \frac{b-a}{n}.$$

We apply the ***Trapezoidal Rule*** on ***each subinterval*** $[x_i, x_{i+1}]$ to approximate the integral:

$$\int_{x_i}^{x_{i+1}} f(x)\,dx \approx \frac{h}{2}\big(f(x_i) + f(x_{i+1})\big).$$

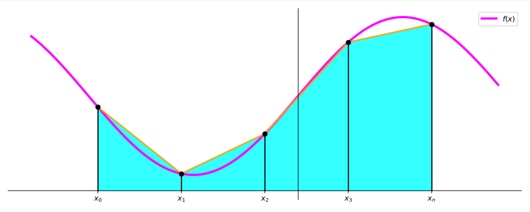
Summing over all ***subintervals***, we obtain

$$\int_a^b f(x)\,dx = \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} f(x)\,dx \approx \frac{h}{2} \sum_{i=0}^{n-1} \big(f(x_i) + f(x_{i+1})\big).$$

After rearranging terms, we obtain

Composite Trapezoidal Rule

$$\int_a^b f(x)\,dx \approx \frac{h}{2}\left(f(a) + 2\sum_{i=1}^{n-1} f(x_i) + f(b)\right).$$



Numerical Integration

Composite Trapezoidal Rule: Example ($n = 4$)

Consider the integral

$$I = \int_0^{\frac{\pi}{2}} \sin(x)\, dx.$$

We use the ***composite trapezoidal rule*** with $n = 4$. Here $h = \frac{b-a}{n} = \frac{\pi/2}{4} = \frac{\pi}{8}$. and $x_i = i\frac{\pi}{8}, \quad i = 0, \ldots, 4$. Thus

$$I \approx \frac{h}{2}\left(f(x_0) + 2\sum_{i=1}^{3} f(x_i) + f(x_4)\right) \approx 0.9871158.$$

Exact value: 1.0000000 Approximation: 0.9871158 Error: 0.0128842

Numerical Integration

Composite Trapezoidal Rule: Example ($n = 8$)

We refine the partition with $n = 8$:

$$h = \frac{\pi/2}{8} = \frac{\pi}{16}, \quad x_i = i\frac{\pi}{16}.$$

$$I \approx \frac{h}{2}\left(f(x_0) + 2\sum_{i=1}^{7} f(x_i) + f(x_8)\right).$$

$$I \approx 0.9967852.$$

Exact value: 1.0000000 Approximation: 0.9967852 Error: 0.0032148

Numerical Integration

Composite Trapezoidal Rule ($n = 4$): Example 2

Consider the integral

$$I = \int_0^{\frac{1}{2}} e^x \sin x \, dx.$$

We apply the ***composite trapezoidal rule*** with $n = 4$: $h = \frac{b-a}{n} = \frac{1}{8}$, $x_i = i\,h$, thus $x_0 = 0$, $x_1 = \frac{1}{8}$,$x_2 = \frac{1}{4}$, $x_3 = \frac{3}{8}$, $x_4 = \frac{1}{2}$, and

$$I \approx \frac{h}{2}\Big(f(x_0) + 2\sum_{i=1}^{3} f(x_i) + f(x_4)\Big).$$

Hence

$$I \approx \frac{1}{16}\Big(0 + 2(e^{1/8}\sin(\tfrac{1}{8}) + e^{1/4}\sin(\tfrac{1}{4}) + e^{3/8}\sin(\tfrac{3}{8})) + e^{1/2}\sin(\tfrac{1}{2})\Big) \approx 0.1743183.$$

$Exact : 0.1716406 \qquad Approx : 0.1743183 \qquad Error : 0.0026777$

Numerical Integration

Composite Trapezoidal Rule ($n = 8$): Example 2

Consider the integral

$$I = \int_0^{\frac{1}{2}} e^x \sin x \, dx.$$

We apply the ***composite trapezoidal rule*** with $n = 8$: $h = \frac{b-a}{n} = \frac{1}{16}$, $\quad x_i = i\,h$. Thus

$$x_i = 0,\ \tfrac{1}{16},\ \tfrac{2}{16},\ \tfrac{3}{16},\ \tfrac{4}{16},\ \tfrac{5}{16},\ \tfrac{6}{16},\ \tfrac{7}{16},\ \tfrac{8}{16}.$$

and

$$I \approx \frac{h}{2}\left(f(x_0) + 2\sum_{i=1}^{7} f(x_i) + f(x_8)\right) I \approx \frac{1}{32}\left(0 + 2\sum_{i=1}^{7} e^{x_i}\sin(x_i) + e^{1/2}\sin(\tfrac{1}{2}\right) \approx 0.1723021.$$

$$\mathit{Exact} : 0.1716406 \qquad \mathit{Approx} : 0.1723021 \qquad \mathit{Error} : 0.0006615$$

Algorithm 1: Composite Trapezoidal Rule

Input : function f, interval $[a, b]$, number of subintervals n

Output : approximation I of $\int_a^b f(x)\,dx$

```
// Step size
```
$h \leftarrow \frac{b-a}{n}$

```
// Initialize sum with endpoints
```
$S \leftarrow f(a) + f(b)$

```
// Accumulate interior contributions
```
for $i = 1, \cdots, n-1$ **do**

  $x_i \leftarrow a + ih$

  $S \leftarrow S + 2f(x_i)$

end

```
// Final approximation
```
$I \leftarrow \frac{h}{2} S$

return I

Numerical Integration

Composite Simpson Rule: Idea

We subdivide the interval $[a, b]$ into n ***subintervals*** with n ***even***:

$$a = x_0 < x_1 < \cdots < x_n = b, \quad x_i = a + ih, \quad h = \frac{b-a}{n}.$$

We apply ***Simpson's Rule*** on ***each pair of subintervals*** $[x_{2i}, x_{2i+2}]$:

$$\int_{x_{2i}}^{x_{2i+2}} f(x)\,dx \approx \frac{h}{3}\big(f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})\big).$$

Summing over all pairs, we obtain

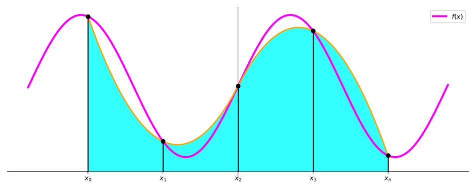
$$\int_a^b f(x)\,dx = \sum_{i=0}^{\frac{n}{2}-1} \int_{x_{2i}}^{x_{2i+2}} f(x)\,dx \approx \frac{h}{3} \sum_{i=0}^{\frac{n}{2}-1} \big(f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})\big).$$

We develop the sum:

$$\sum_{i=0}^{\frac{n}{2}-1} \left(f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})\right)$$

Composite Simpson Rule

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\left(f(a) + 2\sum_{i=1}^{\frac{n}{2}-1} f(x_{2i}) + 4\sum_{i=0}^{\frac{n}{2}-1} f(x_{2i+1}) + f(b)\right).$$



Numerical Integration

Composite Simpson Rule: Example ($n = 4$)

Consider the integral

$$I = \int_0^{\frac{\pi}{2}} \sin(x)\,dx.$$

We use the ***composite Simpson rule*** with $n = 4$: $h = \frac{b-a}{n} = \frac{\pi/2}{4} = \frac{\pi}{8}$. and $x_i = i\frac{\pi}{8}, \quad i = 0, \ldots, 4$. Thus

$$I \approx \frac{h}{3}\Big(f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)\Big) \approx 1.0001346.$$

Exact value: 1.0000000 Approximation: 1.0001346 Error: 0.0001346

We refine the partition with $n = 8$:

$$h = \frac{\pi/2}{8} = \frac{\pi}{16}, \quad x_i = i\frac{\pi}{16}.$$

$$I \approx \frac{h}{3}\Big(f(x_0) + 4\sum_{i=1}^{7} f(x_{2i-1}) + 2\sum_{i=1}^{3} f(x_{2i}) + f(x_8)\Big) \approx 1.0000083.$$

Exact value: 1.0000000 Approximation: 1.0000083 Error: 0.0000083

Numerical Integration

Composite Simpson Rule ($n = 4$): Example 2

Consider the integral

$$I = \int_0^{\frac{1}{2}} e^x \sin x \, dx.$$

We apply the ***composite Simpson rule*** with $n = 4$: $h = \frac{b-a}{n} = \frac{1}{8}$ and $x_i = i\,h$. Thus

$$x_0 = 0,\; x_1 = \tfrac{1}{8},\; x_2 = \tfrac{1}{4},\; x_3 = \tfrac{3}{8},\; x_4 = \tfrac{1}{2},$$

and

$$I \approx \frac{h}{3}\Big(f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)\Big).$$

$$I \approx \frac{1}{24}\Big(0 + 4e^{1/8}\sin(\tfrac{1}{8}) + 2e^{1/4}\sin(\tfrac{1}{4}) + 4e^{3/8}\sin(\tfrac{3}{8}) + e^{1/2}\sin(\tfrac{1}{2})\Big) \approx 0.1716470.$$

$Exact$: 0.1716406 $Approx$: 0.1716470 $Error$: 0.0000064

Consider the integral

$$I = \int_0^{\frac{1}{2}} e^x \sin x \, dx.$$

We apply the ***composite Simpson rule*** with $n = 8$: $h = \frac{b-a}{n} = \frac{1}{16}$ and $x_i = i\,h$. Thus

$$x_i = 0,\ \frac{1}{16},\ \frac{2}{16},\ \frac{3}{16},\ \frac{4}{16},\ \frac{5}{16},\ \frac{6}{16},\ \frac{7}{16},\ \frac{8}{16}.$$

$$\begin{aligned} I &\approx \frac{h}{3}\Big(f(x_0) + 4\sum_{i=1}^{7} f(x_{2i-1}) + 2\sum_{i=1}^{3} f(x_{2i}) + f(x_8)\Big) \\ &\approx \frac{1}{48}\Big(0 + 4\sum_{i=1}^{7} e^{x_{2i-1}} \sin(x_{2i-1}) + 2\sum_{i=1}^{3} e^{x_{2i}} \sin(x_{2i}) + e^{1/2}\sin(\tfrac{1}{2})\Big) \approx 0.1716409. \end{aligned}$$

$Exact$: 0.1716406 $\qquad$ $Approx$: 0.1716409 $\qquad$ $Error$: 0.0000003

Numerical Integration

Comparison of Numerical Integration Methods

Approximation of $\int_0^{\frac{\pi}{2}} \sin(x)\, dx = 1$

Method	n	Approximation	Error
Trapezoidal	1	0.7853982	0.2146018
Trapezoidal	4	0.9871158	0.0128842
Trapezoidal	8	0.9967852	0.0032148
Simpson	2	1.0022799	0.0022799
Simpson	4	1.0001346	0.0001346
Simpson	8	1.0000083	0.0000083
Simpson 3/8	3	1.0010049	0.0010049

Numerical Integration

Comparison of Methods: Example 2

Approximation of $\int_0^{\frac{1}{2}} e^x \sin(x)\, dx \approx 0.1716406$

Method	n	Approximation	Error
Trapezoidal	1	0.1972169	0.0255763
Trapezoidal	4	0.1743183	0.0026777
Trapezoidal	8	0.1723021	0.0006615
Simpson	2	0.1719657	0.0003251
Simpson	4	0.1716470	0.0000064
Simpson	8	0.1716409	0.0000003
Simpson 3/8	3	0.1717693	0.0001287

- Increasing n improves the accuracy
- Higher-order methods (Simpson) are ***more accurate***

Algorithm 2: Composite Simpson Rule

Input : function f, interval $[a, b]$, number of subintervals n (even)

Output : approximation I of $\int_a^b f(x)\,dx$

```
// Step size
```
$h \leftarrow \frac{b-a}{n}$
```
// Initialize sum with endpoints
```
$S \leftarrow f(a) + f(b)$
```
// Add contributions of odd indices
```
for $i = 1, 3, \cdots, n-1$ **do**

$\quad x_i \leftarrow a + ih$

$\quad S \leftarrow S + 4f(x_i)$

end
```
// Add contributions of even indices
```
for $i = 2, 4, \cdots, n-2$ **do**

$\quad x_i \leftarrow a + ih$

$\quad S \leftarrow S + 2f(x_i)$

end
```
// Final approximation
```
$I \leftarrow \frac{h}{3} S$

return I

Numerical Integration

Summary

- Numerical integration consists in approximating

$$\int_a^b f(x)\,dx$$

- Newton–Cotes formulas are based on **polynomial interpolation**:
  - Trapezoidal rule $\rightarrow$ linear approximation
  - Simpson rule $\rightarrow$ quadratic approximation
- Two ways to improve accuracy:
  - Increase the **degree of interpolation**
  - Increase the **number of subintervals** (composite rules)

In practice, **composite Simpson rule** provides excellent accuracy

Refinement is often more efficient than increasing the degree

Numerical Integration

Perspectives: Gaussian Quadrature

- Newton–Cotes formulas use **equidistant points**
- For high degree, they may suffer from **instability** and **loss of accuracy**
- Idea: choose **optimal points** and weights

$$\int_a^b f(x)\,dx \approx \sum_{i=1}^{n} w_i f(x_i)$$

- Gaussian quadrature selects nodes x_i and weights w_i to maximize **accuracy**

☞ It provides **higher accuracy** with fewer points